Sahid Hossain Mustakim

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RESEARCH INTERESTS

Multimodal Deep Learning, Multimodal Adversarial Robustness, Human-AI Interaction, and Explainable & Ethical AI

EDUCATION

TU Dortmund University, Dortmund, Germany — 2026 to Present
Master of Science in Data Science

United International University, Dhaka, Bangladesh — 2020 to 2024
Bachelor of Science in Computer Science and Engineering (**Major:** Data Science) — CGPA: 3.91/4.00
Distinction: Summa Cum Laude — Major CGPA: 4.00/4.00
Thesis Supervisor: Prof. Dr. Swakkhar Shatabda

EXPERIENCE

Part Time Faculty — June 2024–Oct 2026
United International University

- Redesigned the CSE 1110 curriculum with updated concepts and real-world applications.
- Prepared BAETE accreditation documents, mapping course outcomes to IEB-BAETE and Washington Accord standards. Contributed to successful accreditation renewal, valid through December 2029.
- Analyzed student performance and delivered iterative feedback to improve class outcomes.
- Served on the organizing committee of the National CSE Fest 2025 and the UIU CSE Project Show.
- **Courses conducted:** Introduction to Computer Systems, Discrete Mathematics, Structured Programming Language, Structured Programming Language Lab, Object Oriented Programming Lab, Data Structures and Algorithms I & II Lab, Microprocessors and Microcontrollers Lab, Artificial Intelligence Lab, Software Engineering Lab.

Undergraduate Assistant — Oct 2022 – May 2024
United International University

- Designed course materials including assignments, quizzes, and lab exercises across core computer science courses.
- Mentored undergraduate students in programming, data structures, digital logic design, and microprocessor architecture.
- **Courses supervised:** Structured Programming Language Laboratory, Data Structure and Algorithms I Laboratory, Digital Logic Design Laboratory, Microprocessors and Microcontrollers Laboratory, Electronics Laboratory.

PUBLICATIONS

Journals

- Islam, S. M., **Mustakim, S. H.**, Hossain, M., Mashira, M., Shourav, N. I., Ahmed, M. R., Islam, S., Islam, A. K. M., & Shatabda, S. "An audio video-based multi-modal fusion approach for speech emotion recognition." **Accepted, Knowledge-Based Systems (Elsevier)**, Q1, I.F. 7.6, 2026.

Conference and Workshop Papers

- **Mustakim, S. H.**, Islam, S. M. J., Muna, U. M., Chowdhury, M., Islam, M. J., Ahmmed, S., Sikder, T., Dhrubo, S. T. A., and Shatabda, S. "Watch, Listen, Understand, Mislead: Tri-modal Adversarial Attacks on Short Videos for Content Appropriateness Evaluation." Long-paper (Proceedings) track, **SVU Workshop @ ICCV 2025**.
- Islam, S. M. J., **Mustakim, S. H.**, Ahmmed, S., Sayeedi, M. F. A., Khandoker, S., Dhrubo, S. T. A., and Hossain, N. "MIMIC: Multimodal Islamophobic Meme Identification and Classification." 3rd **Muslims in ML Workshop – NeurIPS**, 2024.
- Ahmmed, S., Islam, S. M., **Mustakim, S. H.**, Islam, R., Shanto, S. K., and Islam, S. "IPBlocks: A Blockchain Ecosystem for Secure IP Registration and Decentralized Marketplace." **TENCON 2025 – IEEE Region 10 Conference. IEEE**.

RESEARCH EXPERIENCE

AVLoS: Audio-Visual Long Text Scene Summarization

- **Supervised by**: Dr. Swakkhar Shatabda.
- We propose a multimodal framework for generating fine-grained long-text scene summaries from videos using video and audio inputs. Most approaches use queries to guide text generation, we are experimenting to generate text without these queries.

ChimeraBreak: Coordinated Tri-modal Adversarial Attack on MLLMs

- **Supervised by**: Dr. Swakkhar Shatabda.
- Long paper (Proceedings), SVU Workshop @ ICCV 2025. [paper] [code] [dataset].

- We introduce SVMA, an adversarial dataset for content moderation in short-form videos, and ChimeraBreak. This coordinated strategy exposes systemic safety flaws in leading MLLMs for content appropriateness evaluation. We experiment with open and closed source MLLMs and utilize LLM-as-a-Judge to evaluate ethical reasoning and confidence.

MIMIC: Multimodal Islamophobic Meme Identification and Classification

- **Supervised by**: Nahid Hossain.
- Accepted, MusIML Workshop – NeurIPS 2024. [arXiv] [OpenReview] [code].
- Anti-Muslim hate speech has emerged within memes, characterized by context-dependent and rhetorical messages using text and images that seemingly mimic humor but convey Islamophobic sentiments. This work presents a novel dataset and proposes a classifier based on the Vision-and-Language Transformer (ViLT) specifically tailored to identify anti-Muslim hate within memes by integrating both visual and textual representations.

PROJECTS

NurtureAid [Github]
Developed a real-time AI-based cross-platform mobile application that simplifies care through caretakers.
Tech Stack: React Native, Node.js, Flask, MongoDB, Firebase, LLaMA-Index, PyTorch.
Acknowledgements: Champion of the UIU CSE Project Show, Software Engineering Laboratory; Champion of the Hult Prize OnCampus round in UIU; Selected for the Hult Prize Summit in Boston.

Haven [Github]
Created a BCI-powered mental health speech-to-speech conversational agent that understands and responds with emotion.
Tech Stack: Python, OpenAI, Emotiv, Firebase, Streamlit.
Acknowledgments: Winner, Harvard HSIL Hackathon Dhaka Hub 2024.

BhaShammo [Github]
Solving the unique problem of IPA Transcription of Bengali Regional Dialects using Dialect Guided Tokens.
Tech Stack: Python, Hugging Face, PyTorch.
Acknowledgements: Champion of the Bhashamul national NLP datathon.

ACHIEVEMENTS

Winner, Harvard Health Systems Innovation Lab Hackathon 2025 (Dhaka Hub) April 2025
Academic Excellence Scholarship 2020 — 2024
Champion, Bhashamul: Bengali Regional Text to IPA National NLP Datathon March 2024
Champion, Hult Prize On Campus Round March 2024
Representative, Hult Prize Summit in Boston, USA June 2024
Gold Award (Champion), International Blockchain Olympiad (IBCOL) 2023 November 2023
Silver Award(1st Runner Up), Bangladesh Blockchain Olympiad (BCOLBD) 2023 July 2023
Finalist, ICT Innovation Grant (for NurtureAid) December 2023
Champion, Software Engineering Lab September 2023
3rd Runner Up of the UIU CSE Project Show Sum'22, Database Management Systems Lab May 2022
Champion, Advanced Object Oriented Programming Lab January 2022

SKILLS

Programming: C, C++, Java, Python, JavaScript, PHP, SQL, NoSQL.
Machine Learning: PyTorch, Scikit-Learn, Matplotlib, Numpy, Pandas.
Frameworks/Libraries: React.js, Node.js, React Native, Firebase.
Soft Skills: Communication, Leadership, Teaching, Design Thinking, Public Speaking, Team Player, Content Design.
Tools: MS Word, PowerPoint, Excel, Microsoft Teams, Canva.
DevOps Tools: Git, GitHub.

EXTRA CURRICULARS

UIU Computer Club - served as Vice President Apr 2024 — July 2024
UIU Computer Club - served as Head of Public Relations Jun 2023 — Mar 2024
UIU Computer Club - served as Head of Human Resource Jun 2022 — Jun 2023
UIU Entrepreneur Development Forum - served as Head of Human Resource May 2021 — May 2022

REFERENCES

Dr. Swakkhar Shatabda
Professor of Computer Science and Engineering
Brac University, Dhaka, Bangladesh
E-mail: swakkhar.shatabda@bracu.ac.bd

Md. Rayhan Ahmed
Ph.D. Student, University of British Columbia, Vancouver, Canada & Assistant Professor (on study leave) of Computer Science and Engineering, United International University
E-mail: md.ahmed@ubc.ca